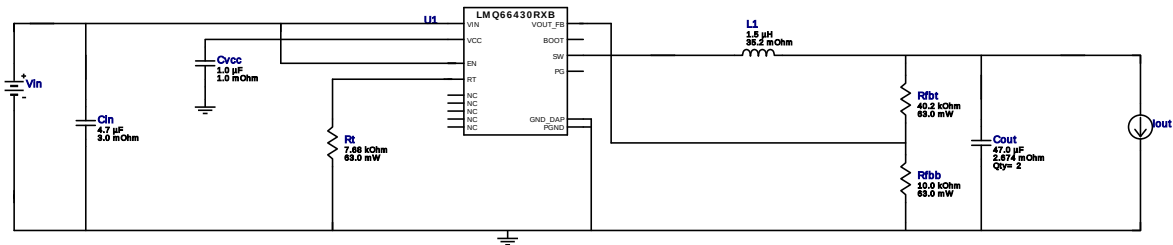


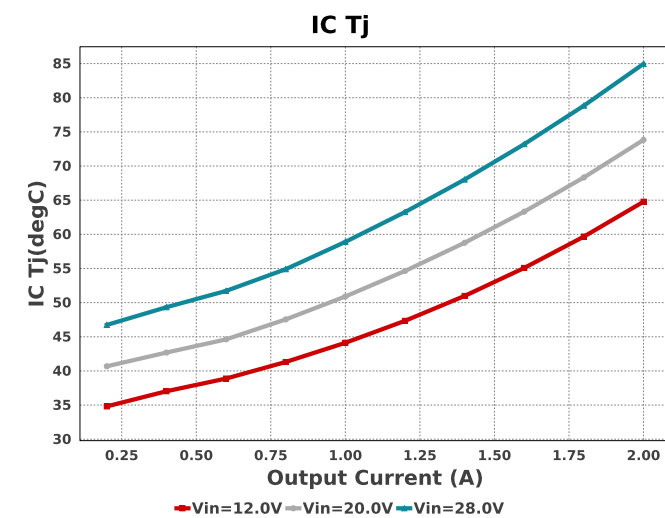
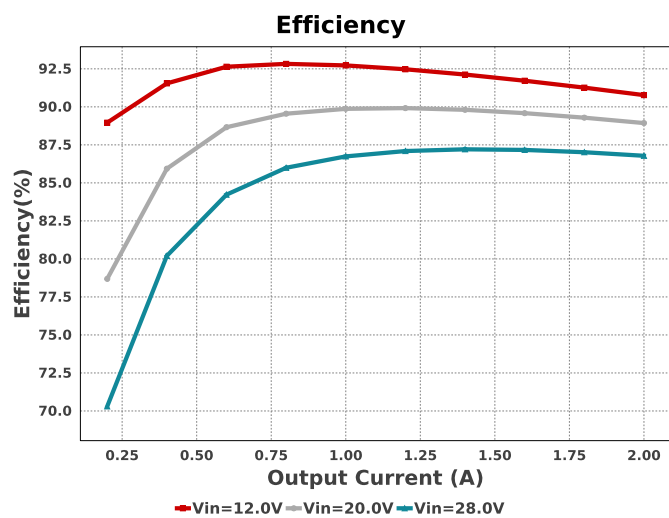
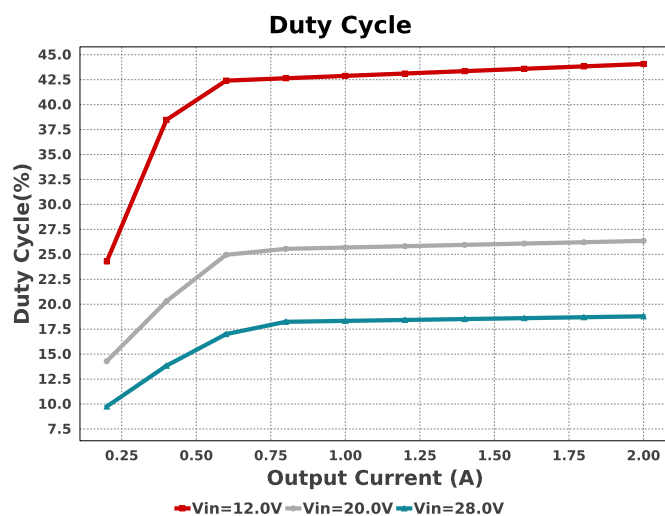
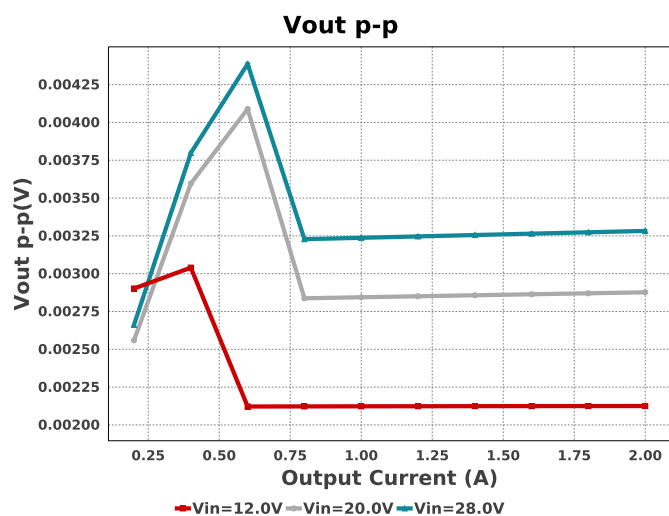
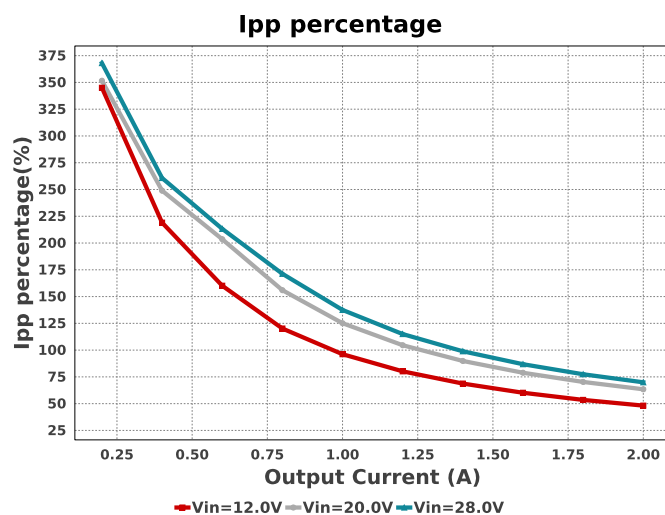
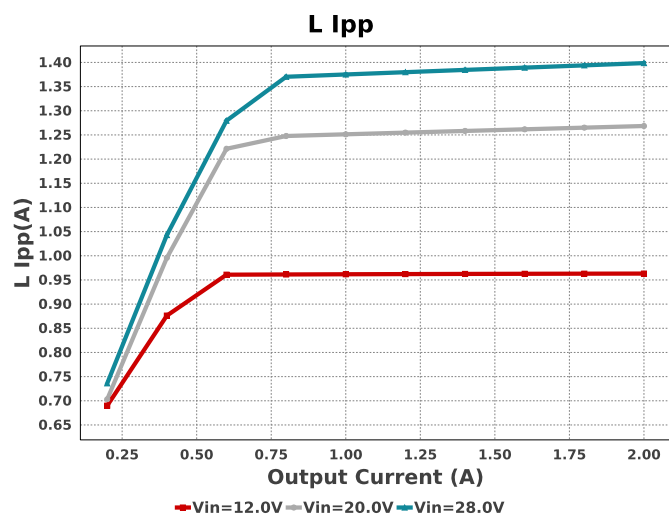
WEBENCH® Design Report

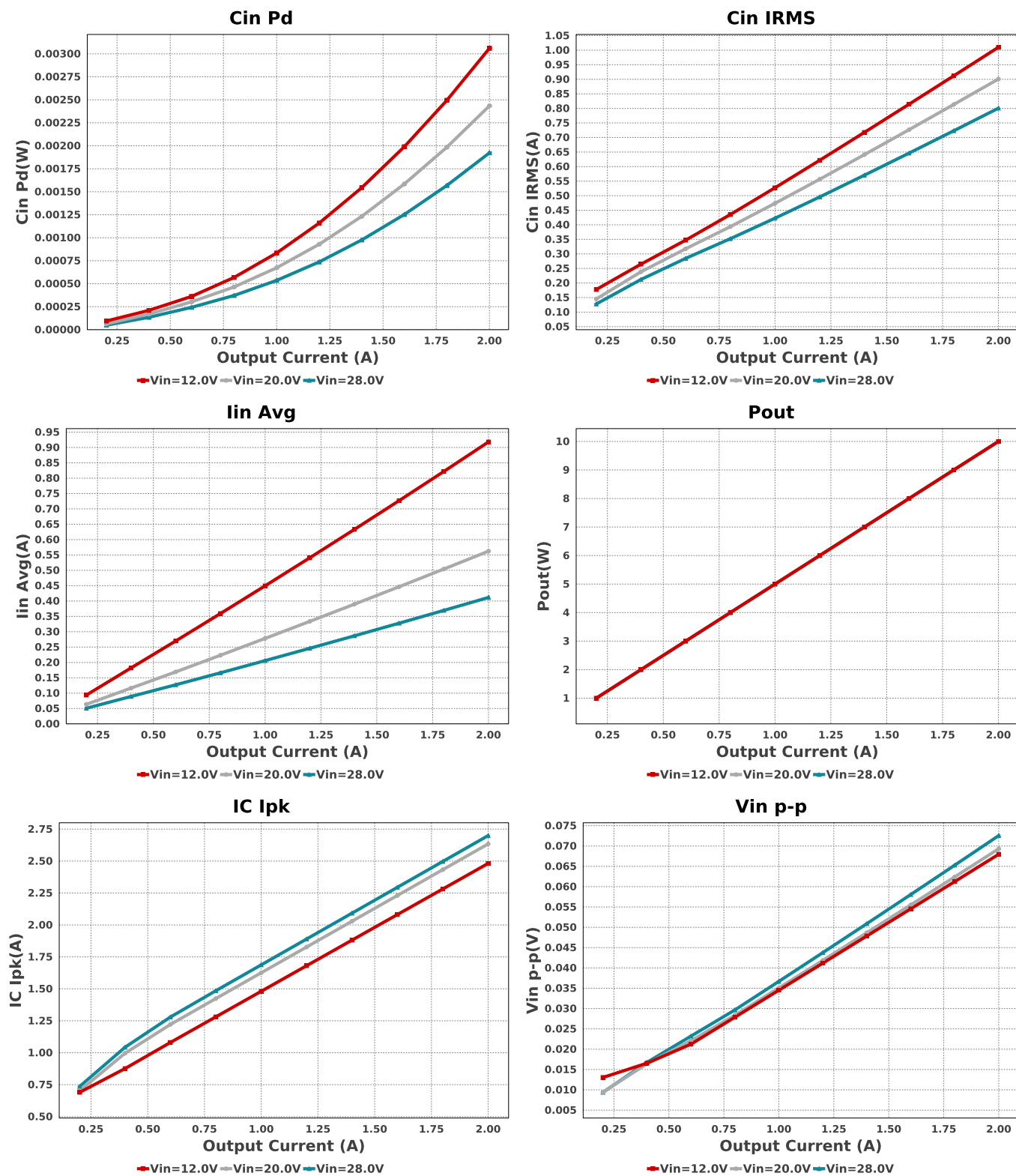
Design : 47 LMQ66430R5RXBR
LMQ66430R5RXBR 13V-28V to 10.00V @ 2A

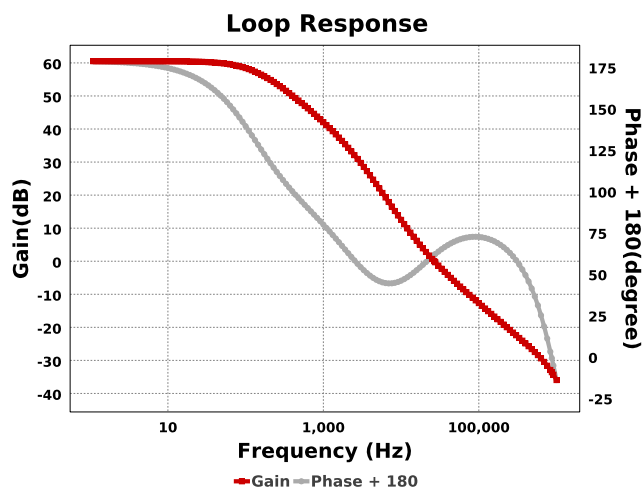
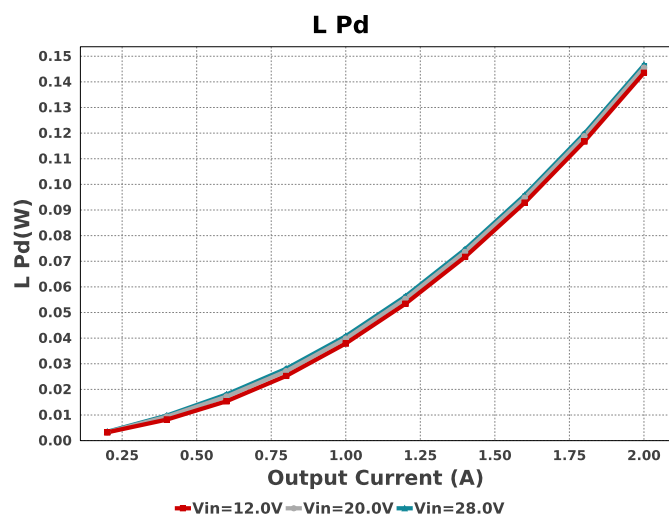
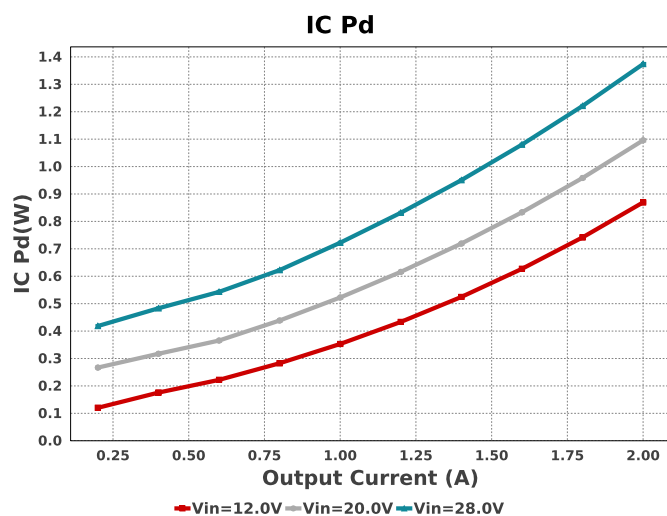
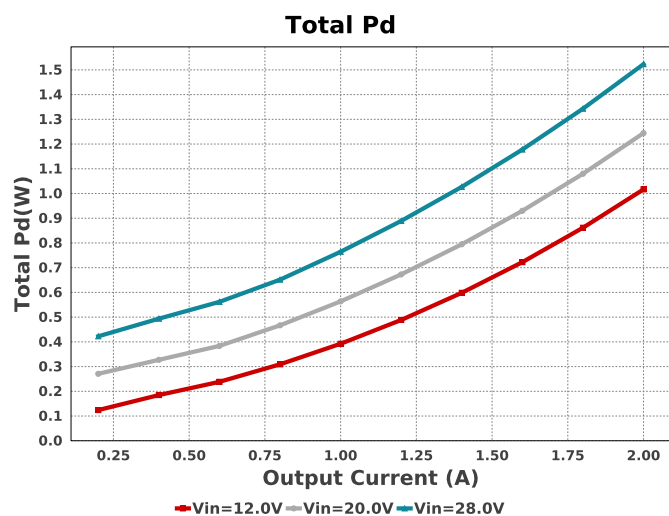
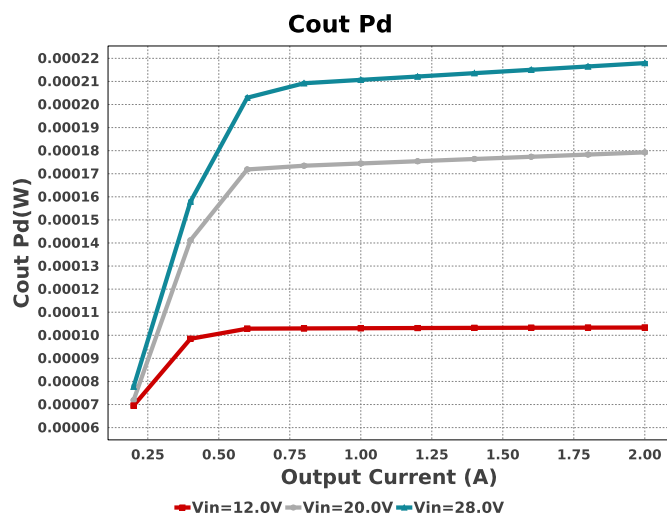
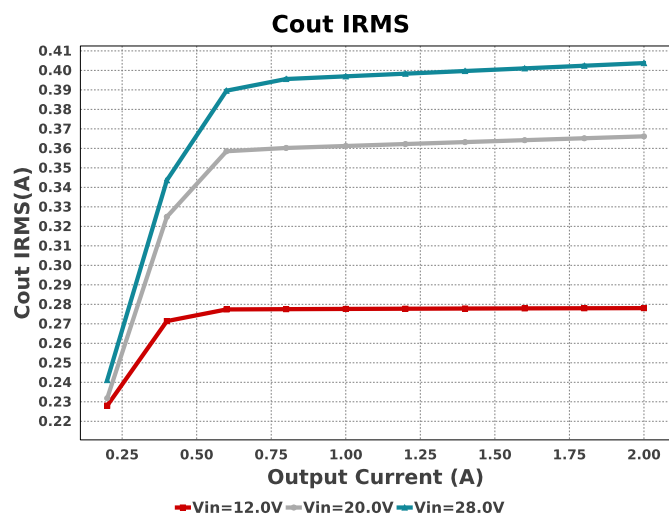


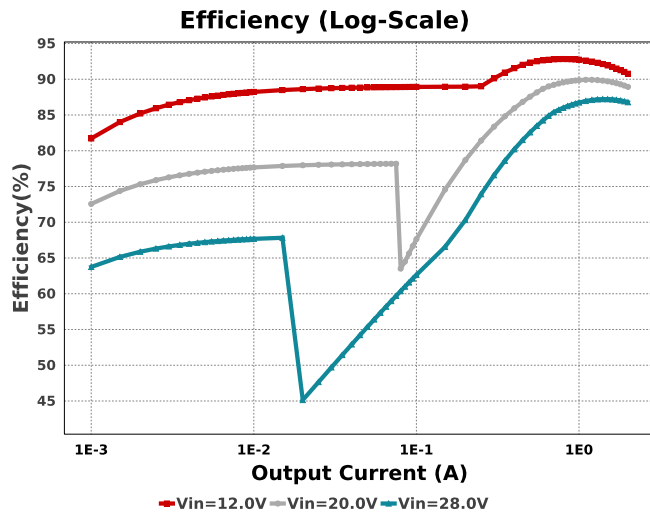
Electrical BOM

Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cin	MuRata	GRM31CR71H475KA12L Series= X7R	Cap= 4.7 uF ESR= 3.0 mOhm VDC= 50.0 V IRMS= 4.98 A	1	\$0.10	 1206 11 mm ²
Cout	Taiyo Yuden	MSASL21GBB5476MTNA01 Series= X5R	Cap= 47.0 uF ESR= 2.674 mOhm VDC= 10.0 V IRMS= 3.1668 A	2	\$0.26	 0805 7 mm ²
Cvcc	Taiyo Yuden	EMK107B7105KA-T Series= X7R	Cap= 1.0 uF ESR= 1.0 mOhm VDC= 16.0 V IRMS= 0.0 A	1	\$0.01	 0603 5 mm ²
L1	Würth Elektronik	74438336015HT	L= 1.5 uH 35.2 mOhm	1	\$1.04	 WE-MAPI_3020 16 mm ²
Rfbb	Vishay-Dale	CRCW040210K0FKED Series= CRCW..e3	Res= 10.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	 0402 3 mm ²
Rfbt	Vishay-Dale	CRCW040240K2FKED Series= CRCW..e3	Res= 40.2 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	 0402 3 mm ²
Rt	Vishay-Dale	CRCW04027K68FKED Series= CRCW..e3	Res= 7.68 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	 0402 3 mm ²
U1	Texas Instruments	LMQ66430R5RXBR	Switcher	1	\$1.42	 RXB0014A 13 mm ²









Operating Values

#	Name	Value	Category	Description
1.	BOM Count	9		Total Design BOM count
2.	Total BOM	\$3.129		Total BOM Cost
3.	Cin IRMS	800.617 mA	Capacitor	Input capacitor RMS ripple current
4.	Cin Pd	1.923 mW	Capacitor	Input capacitor power dissipation
5.	Cout IRMS	403.739 mA	Capacitor	Output capacitor RMS ripple current
6.	Cout Pd	217.94 μ W	Capacitor	Output capacitor power dissipation
7.	IC Ipk	2.699 A	IC	Peak switch current in IC
8.	IC Pd	1.374 W	IC	IC power dissipation
9.	IC Tj	84.958 degC	IC	IC junction temperature
10.	IC Tolerance	12.5 mV	IC	IC Feedback Tolerance
11.	ICThetaJA Effective	40.0 degC/W	IC	Effective IC Junction-to-Ambient Thermal Resistance
12.	Iin Avg	411.54 mA	IC	Average input current
13.	Ipp percentage	69.93 %	Inductor	Inductor ripple current percentage (with respect to average inductor current)
14.	L Ipp	1.399 A	Inductor	Peak-to-peak inductor ripple current
15.	L Pd	146.54 mW	Inductor	Inductor power dissipation
16.	Cin Pd	1.923 mW	Power	Input capacitor power dissipation
17.	Cout Pd	217.94 μ W	Power	Output capacitor power dissipation
18.	IC Pd	1.374 W	Power	IC power dissipation
19.	L Pd	146.54 mW	Power	Inductor power dissipation
20.	Total Pd	1.523 W	Power	Total Power Dissipation
21.	Cross Freq	26.771 kHz	System	Bode plot crossover frequency
22.	Duty Cycle	18.789 %	System Information	Duty cycle
23.	Efficiency	86.782 %	System Information	Steady state efficiency
24.	FootPrint	54.0 mm ²	System Information	Total Foot Print Area of BOM components
25.	Frequency	2.029 MHz	System Information	Switching frequency
26.	Gain Marg	-33.563 dB	System Information	Bode Plot Gain Margin
27.	Iout	2.0 A	System Information	Iout operating point
28.	Low Freq Gain	60.514 dB	System Information	Gain at 1Hz
29.	Mode	CCM	System Information	Conduction Mode
30.	Phase Marg	62.461 deg	System Information	Bode Plot Phase Margin
31.	Pout	10.0 W	System Information	Total output power
32.	Vin	28.0 V	System Information	Vin operating point
33.	Vin p-p	72.559 mV	System Information	Peak-to-peak input voltage
34.	Vout	5.0 V	System Information	Operational Output Voltage
35.	Vout Actual	5.02 V	System Information	Vout Actual calculated based on selected voltage divider resistors

#	Name	Value	Category	Description
36.	Vout Tolerance	2.888 %	System Information	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
37.	Vout p-p	3.283 mV	System Information	Peak-to-peak output ripple voltage

Design Inputs

Name	Value	Description
Iout	2.0	Maximum Output Current
VinMax	28.0	Maximum input voltage
VinMin	12.0	Minimum input voltage
Vout	5.0	Output Voltage
base_pn	LMQ66430R5RXBR	Base Product Number
source	DC	Input Source Type
Ta	30.0	Ambient temperature
UserFsw	2.029 M	Customer Selected Frequency

WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of C_{in} and C_{out} , and the inductance and DC resistance of $L1$ before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

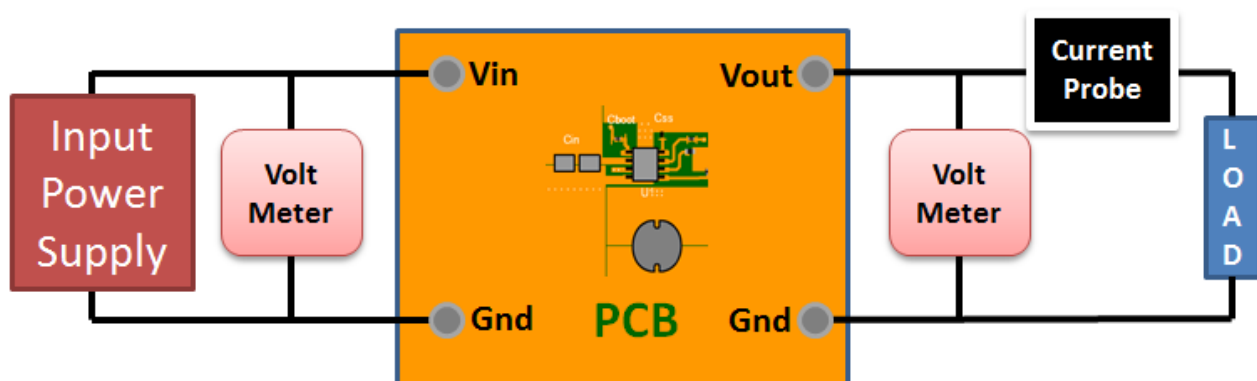
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab down to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 12.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to V_{in} and GND. Connect a digital volt meter and a load if needed to set the minimum load of the design from V_{out} and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between V_{in} and GND, a load is connected between V_{out} and GND and a current meter is connected in series between V_{out} and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



Design Assistance

1. Master key : DC945615436D3D69B25351734F2F9948[v1]
2. **LMQ66430R5RXBR** Product Folder : <http://www.ti.com/product/LMQ66430> : contains the data sheet and other resources.

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